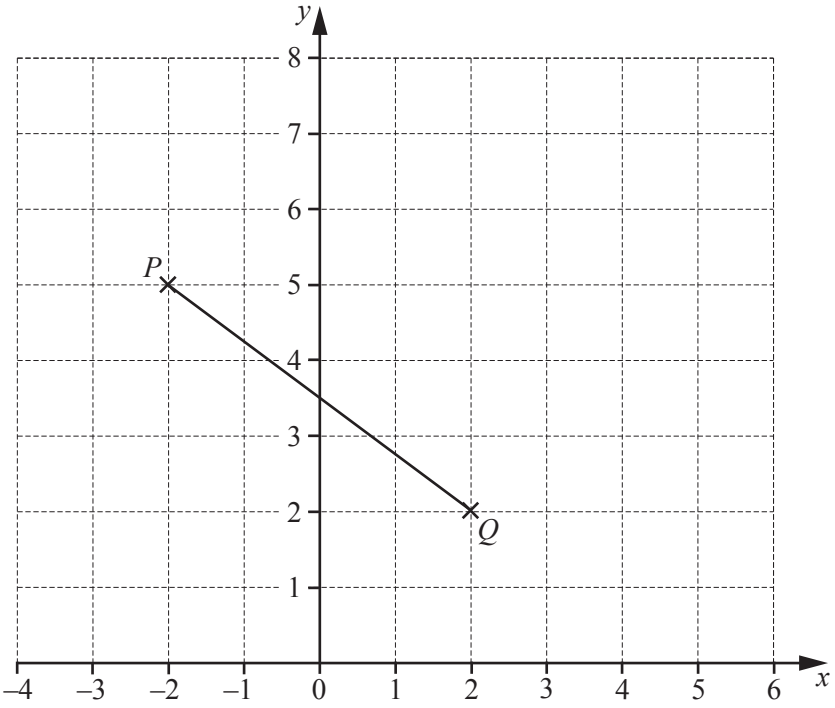
0580_w18_qp_31_Q6

6 (a)



(i) Write down the co-ordinates of point P.

(..... , ...) [1]

(ii) Write down the column vector $\overrightarrow{PQ}$.

$\overrightarrow{PQ} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

(iii) $\overrightarrow{QR} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

On the grid, plot point R. [1]

(iv) $PQRS$ is a parallelogram.

On the grid, complete the parallelogram $PQRS$.
Write down the co-ordinates of point S.

(..... , ...) [2]

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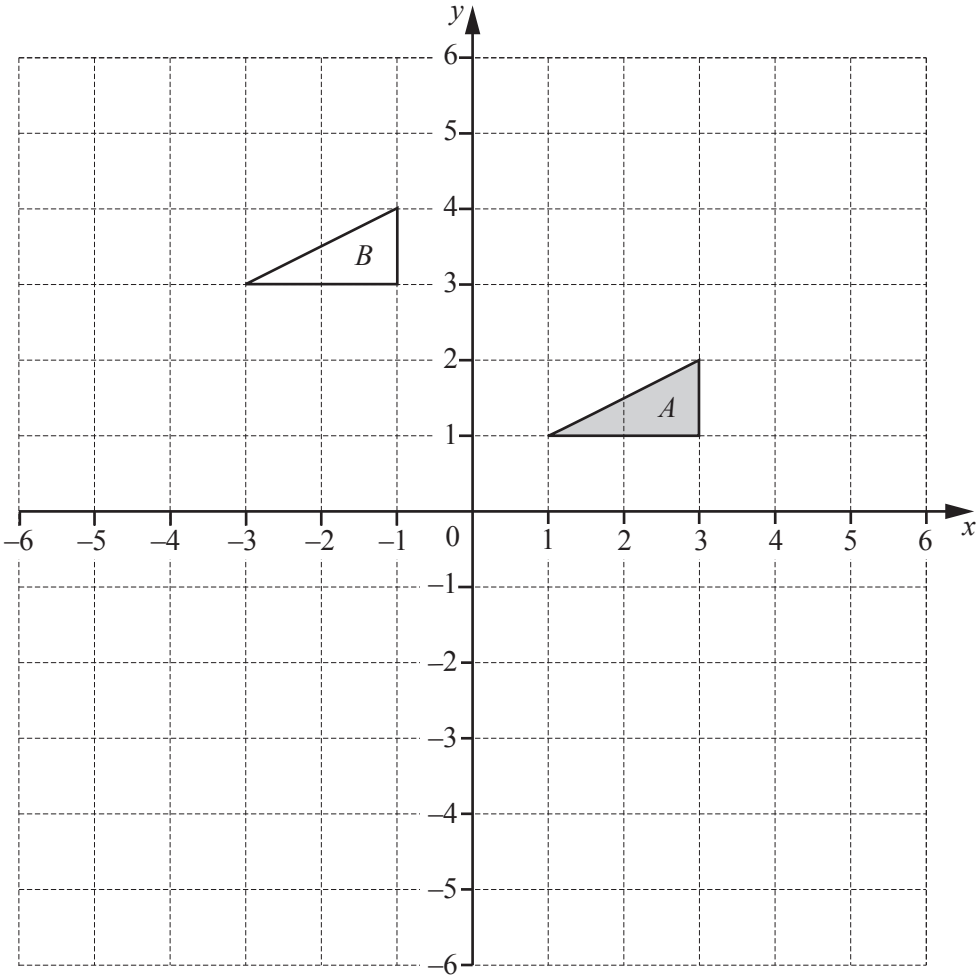
(b)



(i) Describe fully the **single** transformation that maps triangle *A* onto triangle *B*.

.....

..... [2]

(ii) On the grid, draw the image of triangle *A* after a reflection in the line $y = -1$. [2]

(iii) On the grid, draw the image of triangle *A* after a rotation through 180° about (0, 0). [2]